

STEFFAN CHRIST SØLVSTEN

Software Engineer at Scalgo

📍 Aarhus, Denmark [in /steffan-soelvsten](#)



A technophobic computer scientist with a flair for *formal methods* and *algorithm engineering*, and a climber, dancer, swimmer, psychology and philosophy interested and board game playing hippie.

TECHNICAL EXPERIENCE

Software Developer

SCALGO

📅 2019 & May 2026 – Present 📍 Aarhus, Denmark

SCALGO brings cutting-edge massive terrain data-processing technology to market, build on more than two decades of research on I/O-efficient and geometric algorithms.

As a software engineer my responsibilities is to improve and maintain the frontend of the SCALGO Live platform.

Skills

🔧 TypeScript React (S)CSS OpenLayers Git

Postdoctoral Researcher

Aarhus University

📅 April 2025 – March 2026 📍 Aarhus, Denmark

Together with Associate Professor Aslan Askarov, I have worked on extending and improving the information-flow secure and distributed programming language, *Troupe*, to support the implementation of faceted data structures.

📄 github.com/TroupeLang/Troupe

Skills

🔧 Haskell SML TypeScript C

💡 IT Security Information Flow Control Distributed Systems

PhD Student

Aarhus University

📅 November 2019 – February 2025 📍 Aarhus, Denmark

I have, in collaboration with my supervisor, Professor Jaco van de Pol, designed *I/O-efficient* algorithms and data structures to make Binary Decision Diagrams (BDDs) scale beyond the limits of the machine's available memory.

As part of this work, these algorithms have been implemented, evaluated, and improved upon to create a new BDD library in C++, *Adiar*. Compared to conventional implementations of BDDs, our work is thoroughly tested, is faster at scale, and it has a more generic and user-friendly API.

📄 github.com/ssoelvsten/adiar/

📄 ssoelvsten.github.io/adiar/

Skills

🔧 C / C++ Git GitHub Actions

💡 Algorithm Engineering Model Checking I/O-efficiency

EDUCATION

PhD in Computer Science (Datalogi)

Aarhus University

📅 November 2019 – February 2025

MSc in Computer Science (Datalogi)

Aarhus University

📅 August 2019 – August 2022

Master's degree obtained as part of an integrated PhD. My choice of courses focused on *algorithmics* and *formal verification*.

🎓 Course Average: 12.00 (A+).

Skills

🔧 C / C++ Rust SML Rocq

💡 Algorithmics:

- Randomized Algorithms
- Computational Geometry
- I/O-efficiency Complexity Theory

💡 Formal Verification:

- Proof Assistants Separation Logic
- Information Flow Control

BSc in Computer Science (Datalogi)

Aarhus University

📅 August 2015 – June 2018

🎓 Course Average: 11.42 (A).

📄 Bachelor's Project: 12 (A+).

Skills

🔧 Java SML Scheme JavaScript

Python C

💡 Theoretical Computer Science

- Algorithms Data Structures Logic
- Compilation Functional Programming
- Complexity Theory

💡 Mathematics:

- Linear Algebra Mathematical Analysis
- Algebra

Software Developer

IT Minds

March 2018 – March 2019 Aarhus, Denmark

IT Minds provides consultancy to improve and automate the client's workflow. Among my clients have been LEGO, where I was working full stack and was the main architect on the frontend Angular application.

I was also the lead architect on the frontend of an internal project, where I succesfully mentored the new interns, providing feedback on their approaches to solutions and code quality.

Skills

Java

Spring Boot

TypeScript

Angular

React

(S)CSS

Git

MANAGERIAL EXPERIENCE

Supervisor

Aarhus University

Aarhus, Denmark

I have had the pleasure to supervise the following students.

- Sofie Haugaard Olsen and Leonard Hastrup**

Spring 2026MSc Thesis Project
- Hans Christian Hansen**

Spring 2026MSc Thesis Project
- Palle Hauge Petersen**

Spring 2026Talent-track Project
- Erik Funder Carstensen**

Fall 2023MSc Course Project
- Anders Benjamin Clausen and Kent Nielsen**

Spring 2022BSc Thesis Project
- Anna Blume Jakobsen and Mathias Weller Berg Thomasen**

Summer 2020BSc Volunteers

I have also managed the following student programmer.

- Anna Blume Jakobsen**

Spring 2022

Software Developer

IT Minds

March 2018 – March 2019 Aarhus, Denmark

I was the lead architect on the frontend of an internal project, where I succesfully mentored the new interns, providing feedback on their approaches to solutions and code quality.

LANGUAGES

English
Fluent – IELTS Academic: 8.0 (2019)

Danish
Native

German
Native

Peer Review

Aarhus University

Aarhus, Denmark

The academic peer review process requires the ascertainment of the quality of others' piece of work; this also involves providing actionable feedback to the researchers in question.

I have reviewed 1 paper for the following journal.

Information Processing Letters

I have reviewed 5 papers for the following conferences (sorted by research area):

Algorithms and Data Structures

SEA 23

Formal Methods

CONCUR 21 , FMICS 24 , SPIN 24 , TACAS 20

I have reviewed 8 artifacts for the following conferences (sorted by research area):

Algorithms and Data Structures

ALENEX 25–26 †

Formal Methods

ESOP / FASE / FoSSaCS 26 †

† Member of Artifact Evaluation Comittee.

TEACHING EXPERIENCE

Teaching Assistant

Aarhus University

📅 March 2017 – August 2023

📍 Aarhus, Denmark

For a group of students, I corrected their weekly assignments and organized their weekly face-to-face lessons in which they solve the exercises provided by the course coordinator.

- **Software Design using C++** with Bjarne Stroustrup
- **Computability and Logic** with Jaco van de Pol
- **Algorithms and Data Structures** with Kasper Green Larsen
- **Regularity and Automata** with Anders Møller

Substitute Teacher

VUC Fredericia

📅 October 2014 – June 2014

📍 Fredericia, Denmark

I was fully responsible for providing teaching material for and carrying out the lessons in the following three subjects:

- **English** (FED Level)
- **Social Studies** (G Level)
- **History** (FED Level)

I was also supporting other teachers in the following subjects:

- **Danish as a Second Language** (G Level)
- **Mathematics** (FED Level)

Substitute Teacher

10'ende

📅 August 2014 – September 2014

📍 Fredericia, Denmark

I stepped in to cover the first one month of **German** lessons (FED Level). As part of this position, I designed the teaching material myself and carried out the lessons.

OTHER EXPERIENCE

Kitchen Responsible

Regnecentralen, Aarhus University

📅 May 2017 – September 2025

Regnecentralen is a third place for students. I took care of the practical things, organised events, mediated with the university, and created social media content.

Board Member

DSAU, Aarhus University

📅 July 2020 – May 2023

Photographer, videographer, graphics artists, and everything else related to DSAU's online and offline media presence.

Bartender

Fredagscaféen, Aarhus University

📅 May 2017 – December 2019

Computer Science's very own "Fredagsbar" at Aarhus University. I have been bartending two to four times every semester.

PUBLIC SPEAKING

I have made technical research results in computer science palpable as part of the talks listed below.

Talks marked with 📺 have been recorded; links to the videos are available in the list of publications below.

VMCAI [2] 📺 📅 January, 2026
PriSC [8] 📺 📅 January, 2026

SPIN [3] 📅 April, 2024

ATVA [4] 📅 October, 2023
NFM [5] 📅 May, 2023

MOVEP 📅 June, 2022
TACAS [6] 📅 April, 2022

MFCS [7] 📺 📅 August, 2020

REFERENCES

Jaco van de Pol

@ Aarhus University

✉ jaco@cs.au.dk

PhD Supervisor [1, 3–6]

Kristoffer Arnsfelt Hansen

@ Aarhus University

✉ arnsfelt@cs.au.dk

Mentor for a project in game theory [7]

Theatrical Technician

TÅGEKAMMERETs Revy, Aarhus University

📅 December 2021 – December 2024

I joined on short notice to livestream the revue. Since then, I have taken care of the camera and more at the live shows.

Onboarding Mentor

Mat/Fys-Tutorgruppen, Aarhus University

📅 January 2016 – December 2017

Planning and delivering a warm welcome first years for their first semester at the University. This included both social and university related questions.

Being in the LaTeX group, I have been the main responsible to completely redo from scratch all code producing the layout and design of their yearly songbook.

PUBLICATIONS

PhD Thesis

1. Steffan Christ Sølvsten
“**I/O-efficient Symbolic Model Checking**”.
In: *Royal Library, Denmark*. 2025.
https://soeg.kb.dk/permalink/45KBDK_KGL/1pioq0f/alma99126389524805763

Conference Proceedings

Unlike many other areas of research, computer scientists primarily publish their research results in *conference proceedings* rather than *journals*. This is not at the cost of quality of the research since these publications are thoroughly peer reviewed. Similar to journals, conferences are *ranked*, e.g. our publication at TACAS [6] is at an A-tier conferences.

2. Steffan Christ Sølvsten and Jaco van de Pol.
“**Multi-variable Quantification of BDDs in External Memory using Nested Sweeping**”.
In: *Verification, Model Checking, and Abstract Interpretation (VMCAI)*. 2026.
doi:10.1007/978-3-032-15700-3_17 youtu.be/WKFXUsd5oqY

3. Steffan Christ Sølvsten, Casper Moldrup Rysgaard, and Jaco van de Pol.
“**Random Access on Narrow Decision Diagrams in External Memory**”.
In: *International Symposium on Model Checking Software (SPIN)*. 2024.
doi:10.1007/978-3-031-66149-5_7

4. Steffan Christ Sølvsten and Jaco van de Pol.
“**Predicting Memory Demands of BDD Operations using Maximum Graph Cuts**”.
In: *Automated Technology for Verification and Analysis (ATVA)*. 2023.
doi:10.1007/978-3-031-45332-8_4

5. Steffan Christ Sølvsten and Jaco van de Pol.
“**Adiar 1.1: Zero-suppressed Decision Diagrams in External Memory**”.
In: *NASA Formal Methods (NFM)*. 2023.
doi:10.1007/978-3-031-33170-1_28

6. Steffan Christ Sølvsten, Jaco van de Pol, Anna Blume Jakobsen, and Mathias Weller Berg Thomasen.
“**Adiar: Binary Decision Diagrams in External Memory**”.
In: *Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*. 2022.
doi:10.1007/978-3-030-99527-0_16

7. Kristoffer Arnsfelt Hansen and Steffan Christ Sølvsten.
“ **$\exists$ R-Completeness of Stationary Nash Equilibria in Perfect Information Stochastic Games**”.
In: *Mathematical Foundations of Computer Science (MFCS)*. 2020.
doi:10.4230/LIPIcs.MFCS.2020.45 youtu.be/CXC2UMi6hg0

Workshops

8. Steffan Christ Sølvsten and Aslan Askarov.
“**Efficient Dependency Resolution in IFC-aware Decentralized Programming**”.
In: *Principles of Secure Compilation (PriSC)*. 2026.
youtu.be/1VPPJ8uUXdE

Preprints

9. Steffan Christ Sølvsten and Jaco van de Pol.
“**Symbolic Model Checking in External Memory**”.
2025.
doi:10.48550/arXiv.2505.11229